

Gain A New Perspective: Towards Exploring Multi-View Alignment in Mixed Reality

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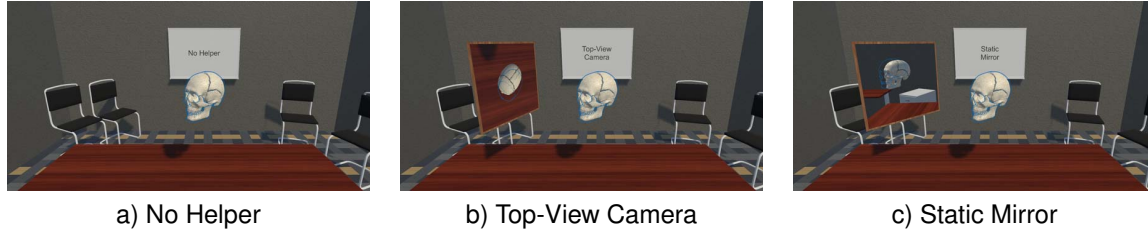


Figure 1: Examples of virtual helpers provided to assist users during alignment tasks. Apparent correct alignment can be perceived (a) when no virtual helper is present. The use of virtual helpers such as (b) camera views, and (c) virtual mirrors, provide additional information to identify misalignment errors.

ABSTRACT

Manufacturing, maintenance, assembly, and training tasks represent some of the human activities that have captured special interest for Mixed Reality (MR) applications. For most of these scenarios, accurate object alignment constitutes a requirement to ensure the desired outcome. This task has proved to be especially challenging in egocentric approaches, frequently leading to estimation errors in depth. While traditional MR methods provide virtual guides such as text, arrows, or animations to assist users during alignment, this work explores the feasibility of using additional views generated by virtual cameras and mirrors. Presenting additional views from different perspectives can help to mitigate the estimation errors and show information that is not directly visible to users.

To explore the benefits of using additional views for alignment tasks, and to collect reliable data and diminish external factors, we conducted a user study in a controlled virtual environment where participants aligned objects supported by additional views from a top-down camera and virtual mirrors. Data regarding alignment error, time to completion, user's interaction and attention, distance traveled, average head velocity, usability, and mental effort were collected. Our results show that using additional views reduces the mental effort and distance traveled by users and increases acceptance without negatively affecting the alignment accuracy. Therefore, we believe that users will also benefit from integrating these techniques during alignment tasks in MR environments.

Index Terms: Human-centered computing [Human computer interaction (HCI)]; HCI design and evaluation methods—User studies; Human-centered computing [Human computer interaction (HCI)]; Interaction paradigms—Virtual reality; Human-centered computing [Visualization]; Visualization design and evaluation methods—

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1 INTRODUCTION

Manufacturing, maintenance, and training are some of the human activity areas that have captured special interest for Virtual, Augmented and Mixed Reality applications since their foundation, *hereinafter referred as VR, MR and AR*. Existing solutions have shown the advantages of using these technologies for training, manufacturing, and assembly in industrial environments [4, 6, 9, 27, 29].

Achieving accurate alignment and placement of virtual and real objects represents a crucial action performed by users in several spatially aware AR solutions in industrial and medical applications. This task has proven to be specially challenging when using egocentric approaches due to the lack of information along the line of sight from the user's head, frequently leading to alignment errors in depth [14]. To solve this problem users frequently change their position to obtain a different view, however, this leads to an iterative process in which improving alignment in one view can affect the result of the previous iteration.

While traditional AR approaches present virtual content in the form of text, arrows, animations, or virtual replicas of the real objects to assist users during alignment, they also require knowing the pose of the objects to be aligned. Conversely, our work explores the feasibility of using multi-view approaches for alignment in Mixed Reality scenarios where no tracking of these objects is available. In this case, the spatial relation between the desired and actual poses of the objects is unknown. Therefore, the effectiveness of the alignment task depends on the accuracy that can be achieved by the user's interactive alignment without the assistance of any guidance. This scenario illustrates a common setup for multiple MR applications that has remained a challenging and open topic.

In this regard, we believe that presenting additional views can help in resolving depth ambiguities that are present in egocentric AR and VR systems, as they provide visual information that can be used to improve and support the successful completion of the alignment task. Moreover, presenting multiple views can benefit the alignment task in cluttered scenarios in which the user's movements are restricted as it is the case of several industrial and medical scenarios. As depicted in Fig. 1, users can perceive correct alignment when using a single view approach (Fig. 1a), however, the presence of multiple views can help to identify existing misalignment (Fig. 1b and Fig. 1c).

Even though the use of multiple viewpoints have been explored in the past: to provide alternative views of single or multiple scenes in VR [11], to generate virtual mirrors to improve perception in the

context of laparoscopy [1], to avoid user's displacement in tasks that require multiple viewpoints [32], or to generate reflective windows to assist during the setup of robotic arms in minimally invasive surgery [7]; in this work we present a thorough evaluation of the user's performance when using multiple views for alignment tasks.

As the first step towards exploring the benefits of using additional views during object alignment in AR, and to collect reliable information and diminish external factors such as inaccurate tracking results, or drift effects associated to commercially available AR optical see-through head-mounted displays, the evaluation of our solution was conducted in a virtual environment. Users' performance and behavior were analyzed while interacting with three different *virtual helpers*: 1) a top-down view from a virtual camera placed on top of the alignment area (Fig. 1b), 2) a static virtual mirror able to only adjust its orientation to ensure optimal visualization of the alignment area (Fig. 1c), and 3) a dynamic virtual mirror that automatically adjusts both, position and orientation based on the user's viewpoint.

Data collected from the objects' pose and time to completion were used to evaluate performance during alignment. Users' interactions were computed based on the manipulation events occurred per trial. Acquired eye tracking data served to characterize interaction with the virtual helpers as well as to recognize the object's areas where users focused their gaze during alignment. Additionally, data collected from the head pose was used to compute the users' distance traveled, as well as their average head velocity. After task completion, usability and perceived mental effort questionnaires were collected to identify the complexity of using the proposed techniques.

A review of related work is presented in Section 2. The technical considerations for the design of the virtual helpers are described in Section 3. Section 4 contains details about the experiment design, the implementation of our study as well as the data collection, and Section 5 presents a statistical analysis of the data acquired. Further discussion based on the statistical outcomes and our conclusions are presented in Section 6 and Section 7, respectively.

2 RELATED WORK

Evaluating the feasibility of AR-based systems for manufacturing, maintenance, and training applications in industrial environments has been a widely explored topic. Caudel and Mizell [4] showed the benefits of using AR for manufacturing in the aerospace industry. The KARMA project proposed by Feiner et al. [6] provided printer maintenance instructions to end-user using head-mounted displays. Reiners et al. [29] developed an AR-based system to provide animated instructions for door assembly in automotive scenarios. Henderson et al. [9] evaluated the performance of using AR when compared with traditional learning methods during the execution of procedural tasks. Pringle et al. [27] introduced a system able to provide instruction for novice users during motor maintenance in real-life industrial scenarios.

Accurate alignment of real objects is not limited to industrial settings. For instance, in the medical context Leuze et al. [21] presented a system able to guide physicians during the placement of electromagnetic coils for the treatment of depression. This system shows a virtual replica of the medical device next to the patient's head indicating the optimal position on which the coil must be placed. Achieving accurate alignment of the device improves the efficiency of the treatment. Unberath et al. [35] presented a system capable of recording desired poses from C-Arm imaging devices to assist X-Ray technicians during surgery. The virtual C-Arm at these planned poses was visualized by the technicians using an AR HMD. In this scenario, achieving the proper alignment between the real C-Arm scanner and the desired virtual configuration resulted in fewer shots, hence reduced the overall radiation exposure.

Alternative solutions used generic models such as interlocking plastic bricks to demonstrate the usability of AR for assembly tasks. Tang et al. [34] evaluated the effectiveness of using AR for ob-

ject assembly when compared against common instructional media, suggesting that using AR improves task performance and relieves mental workload. Khuong et al. [17] used different visualizations to guide users during assembly tasks, showing that side-by-side techniques are preferred when compared with direct object overlaying on top of physical models. Robertson et al. [30] evaluated the effects of presenting virtual content located outside of the task area during assembly, showing that using non-registered AR does not present performance improvements over heads-up instructions, however, is considered less intrusive. These observations reaffirm the importance of intuitive registration and alignment between virtual and real content in an AR environment.

Presenting alternative views during alignment provides meaningful visual information to overcome the underestimation of depth observed when egocentric systems are used in AR and VR, especially when environmental spatial constraints exist. These underestimation effects have been reported by Loomis and Knapp [22], Willemsen et al. [37], and Creem-Regehr et al. [5] in VR environments; and by Hoff and Vincent [14], Swan et al. [33], and Jones et al. [16] in AR.

Martin-Gomez et al. [23] presented a conceptual evaluation of visualization techniques used for object alignment in VR. The selection and classification of the proposed techniques were based on the level of occlusion observed when the objects overlapped during task performance. Results obtained from this study suggest that techniques with reduced level of occlusion increase user's preference and acceptance without compromising accuracy.

The advantages of using multiple viewpoints in AR and VR has been explored in the past. Stoakley et al. [31] introduced the *worlds in miniature* metaphor that offered multiple virtual views to users by creating a miniature replica of a virtual environment. This system allowed users to manipulate the virtual environment using the miniature. Results from this study showed that users can simultaneously manipulate objects while changing their points of view. The work presented by Kiyokawa and Takemura [18] introduced the concept of a *tunnel window* able to provide multiple viewpoints in VR. The proposed windows allowed users to navigate and interact with content in single or multiple scenes. Nakamura et al. [25] provided users with a third-person view for games in VR. In this experiment, the video sequence acquired from an external camera was visualized by the users using an HMD. Results from this study showed that users adapt fast to these views. Nevertheless, when users ended facing towards the external camera, the mirrored view produced cognitive difficulties. Hoang and Thomas [12, 13] used external cameras to generate an *augmented viewport* that was visualized in a Picture-in-Picture style in combination with the user's direct view for AR outdoor applications. By using the viewport for the manipulation and alignment of virtual objects on top of buildings, users required less time and effort, and improved precision during task performance. Alternatively, Sukan et al. [32] presented a scenario where virtual objects were aligned given pairs of real objects in AR. This system used pre-acquired photographs from different viewpoints to assist users during the alignment tasks. Results from this study showed that users can achieve faster alignment using this system when compared to scenarios where users had to walk. Moreover, the work presented by Grandi et al. [8] utilized hand-held devices to enable simultaneous collaborative manipulation of virtual objects in AR.

Furthermore, the use of multiple views has also been used for collaborative environments. The Dollhouse system presented by Ibayashi et al. [15] used a VR approach in the context architectural design of living or working spaces. In this work, a top-down view of a floor-plan allowed non-immersed users to provide details regarding the environment to a user immersed in the virtual world. This system allowed communication between the users by allowing the immersed users to see through the ceiling of the virtual environment as well as to enabled pointing targets from both types of users. A similar approach presented by Kunnert et al. [19] used a tabletop and a



Figure 2: The handle (Top) and target (Bottom) objects used to evaluate users' performance during alignment. From left to right: Mug, Camera, Skull, and Balloon.

vertical screen to present egocentric and allocentric views of a virtual environment. These views were used to provide distinguishable navigation capabilities during scene exploration tasks.

Other solutions have used mirrored-like views to improve users' experience in AR applications. Bichlmeier et al. [1] proposed the use of *virtual mirrors* as an interaction paradigm to augment the direct view of physicians in laparoscopic applications, providing additional views from volumetric medical data without the need of moving the patient. Fotouhi et al. [7], introduced the concept of AR-reflective displays to improve the workflow when setting up robotic arms used for trocar placement in minimally invasive surgery. This concept used pre-acquired images from multiple viewpoints of the scene that were inverted over its horizontal axis to provide a quasi-mirrored view from their respective viewpoints. Despite that these displays were not able to reflect changes in the real content, results from this study indicated an improvement when registering a virtual replica of the robot with its real counterpart.

3 METHODS

To evaluate the effects of providing additional views during object alignment in AR, we conducted a study in a controlled virtual setup. This allowed us to design a controlled environment in which the geometrical properties of the objects were identical, and contributed to collect reliable data and diminish external factors such as system-related errors commonly associated to AR setups that may result in much higher errors than the ones committed by the user during alignment. Examples of these factors are imprecise eye to display calibration, split attention problems due to reduced field of view (as the user can struggle to visualize the object of interest and the virtual helper simultaneously), the lack of sufficiently accurate interaction methods in 3D as they are either imprecise or cumbersome to use (e.g. hand gestures and air taps), or inaccurate tracking results or drift effects observed when using commercially available AR optical see-through head mounted displays. A similar approach was presented by Lee et al. [20], where a VR environment was used to replicate an AR system.¹

Participants of our study were asked to align textured virtual objects—hereinafter referred to as *handle objects*—with respect to a pose defined by their virtual replica—hereinafter referred to as *target objects*—. Our system incorporates direct manipulation techniques for the interaction during task performance based on the results from the studies conducted by Mine et al. [24] in VR.²

Quantitative scores corresponding to alignment accuracy in position and orientation, time to completion, user's interaction, eye gaze, distance traveled, and average head velocity were collected. Posterior evaluation questionnaires served to collect qualitative data regarding usability and mental effort.

¹Results from this study suggest that VR simulations of AR environments can be a viable alternative towards performing controlled experiments.

²This work explored how body-relative interactions affect user performance during object docking tasks.

3.1 Models Selection

To obtain data comparable with existing results for object alignment and attain a baseline for our comparison, the models used during the alignment tasks in this work correspond to those used by [23]. These models are shown in Fig. 2 and contain different geometrical properties that can be used by the participants during alignment. The *Mug* has a curved homogeneous surface, with a handle containing the most dominant structure for achieving the alignment. The *Camera* presents a combination of planar and curved surfaces from which, the lens represent one of the most significant shape cues. The *Skull model* does not contain notorious geometrical extensions that can provide useful cues for orientation, however, the sharp edges observed in the mandible and frontal bones, can be used to achieve the task. Lastly, the extended extremities of the *Balloon* are useful to perceive possible orientation misalignment.

3.2 Target Objects Visualization

The selection of the visualization technique used to represent the *target objects* corresponds to the one that achieved the best overall results reported by [23]. This technique highlights the model's borders by generating an outline based on the view direction of the users and the surface normal of the model. As the outline visualization grows outwards the model surface, the *handle object* is able to fit inside of its respective *target object* without generating significant occlusion to the user when the objects overlap (see Fig. 2).³

3.3 Virtual Helpers

To provide additional and meaningful information to the users and overcome the limitations associated with scenarios where physical restrictions in the environment limit the user movements, we propose to augment the environment by presenting additional views. By presenting views from different perspectives using virtual cameras and mirrors, we expect to simplify the alignment task because these types of views are familiar to humans. A tangible example this is how humans use multiple mirrors to align a car with the curb while parking their cars. Furthermore, we expect that providing this additional information will positively affect performance during alignment tasks.

3.3.1 Top-View Camera

This helper presents an image generated by a virtual camera placed on top of the *target object*. The virtual camera with a field of view of 30° was placed 1 meter above the target. While presenting the additional view orthogonal to the user's viewpoint would provide the highest amount of depth information, positioning the virtual helper too far away from the user's view can introduce undesired split attention artifacts. Therefore, the image generated was always shown at 0.5 meters from the target object and forming an angle of 150° between the World-Right and Target-to-Helper vectors as depicted in Fig. 3a and Fig. 3b. An example of the image rendered with this technique is shown in Fig. 1b. This type of top-down views is common in clinical settings where cameras and scanners are used to provide useful images inside the operating room.

3.3.2 Static Mirror

Conversely to the *Top-View Camera*, the *Static Mirror* renders an image simulating a mirrored reflection accompanied with parallax properties instead of a projection. This reflection corresponds to the image that is observed when looking in a real mirror. In this regard, any changes on the user's pose will produce changes in the texture observed on the mirror surface. The position of the mirror corresponds to the one used for the image shown by the

³The Mug, Camera and Skull models are available at: <https://free3d.com/>; The Balloon is available at: <https://docs.unity3d.com/Manual/SL-VertexFragmentShaderExamples.html>

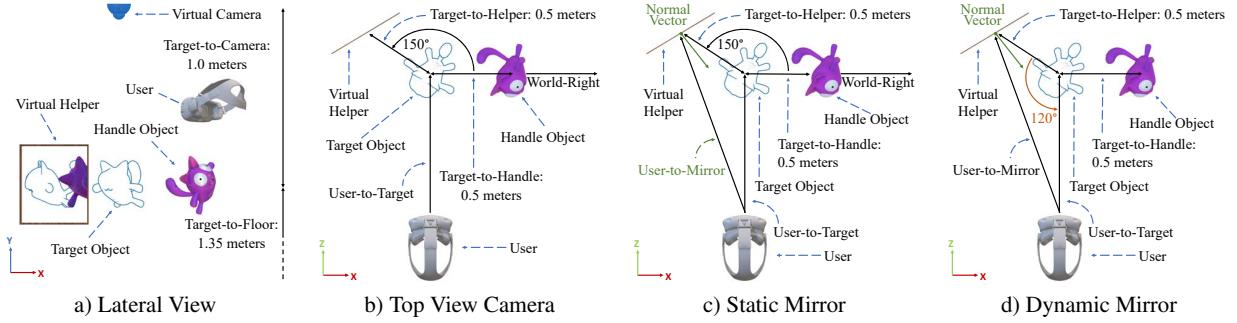


Figure 3: Distribution and placement of the virtual objects and helpers on the scene. The lateral view (a), is the same for all the virtual helpers. The virtual helper is static in position and orientation for the *Top View Camera* (b), static in position and dynamic in orientation for the *Static Mirror* (c), and dynamic in position and orientation for the *Dynamic Mirror*.

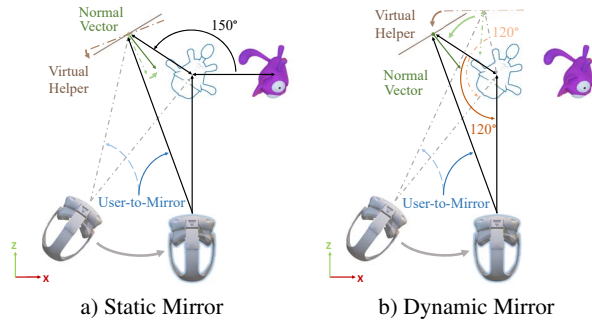


Figure 4: Distribution and placement of the virtual objects and helpers when the user moves during the alignment trial. The *Static Mirror* modifies its orientation to ensure that the target object is always visible (a), while the *Dynamic Mirror* additionally changes its position to ensure an angle of 120° from the user (b).

top-view camera, however, its orientation is updated continuously to ensure that the target object is always visible, ensuring that the image shown is comparable to the one obtained by the top-view camera. To achieve this, the normal vector of the mirror was set to be facing towards the middle angle defined by the Mirror-to-User and Mirror-to-Target vectors (see Fig. 3c and Fig. 4a). An exemplary image resulting from this helper can be seen in Fig. 1c.

3.3.3 Dynamic Mirror

Lastly, we propose the use of a *Dynamic Mirror*. This mirror differs from the static version in the sense that both, position, and orientation are dynamically updated based in the position of the user on the room. The orientation of the mirror is computed the same way as the static version, and its position is always at 0.5 meters from the Target and forming a 120° angle between the User-to-Target and Target-to-Helper vectors (as depicted in Fig. 3d and Fig. 4b).

4 USER STUDY

To evaluate the effects of augmenting the environment with additional viewpoints during alignment tasks, we developed a virtual application using the HTC Vive Pro Eye headset and Unity 3D. A pilot study served to define, validate, and refine the initial design considerations of the system proposed. A subsequent user study was conducted in a private office, where external distractions were reduced, to collect quantitative and qualitative data used to evaluate the performance of the virtual helpers.

Our experimental setup is composed of two stages: i) a tutorial and ii) an evaluation scenario. Data regarding accuracy in position and orientation, time to completion, user's interaction, focus of attention, distance traveled, and average head velocity were collected. After completion of the experiments, users were asked to answer an evaluation questionnaire for usability and mental effort. The total

duration of the experiment was between 25 and 40 minutes. More details regarding the design of our system and the user study are described in the remaining of this Section.

4.1 Participants

A total of thirty-two users (9 female and 23 male), all of them unpaid volunteers aged between 21 and 46 with a mean age of 26.8 ± 4.6 years gave their consent to take part in the study. Over the 32 participants, 5 reported using AR/VR systems at least once a month and 23 never used the HTC Vive system before.

4.2 Experimental Setup

Participants were provided with the HTC Vive headset and asked to perform an eye tracking calibration routine, including the adjustment of the headset and the interpupillary distance. All participants wearing glasses to correct vision problems were able to use them during the experiment. After device calibration, an introductory screen presented our system to the users. To emulate those scenarios in which users' movements are limited by the environment, a real desk was placed in the evaluation room and represented in the virtual setup. The desk was positioned in front of the users while the virtual objects and helpers were placed above. Users were allowed to move laterally but instructed to avoid moving around the desk.

4.2.1 Interaction Modes

Two interaction modes were available for object manipulation: *Normal mode* allowed users to move the handle object by directly attaching it to the device controller. *Precise mode* reduced the translation and orientation changes of the handle object by a ratio of one tenth with respect to the changes observed in the controller's pose. The precise mode was enabled exclusively when the distance between the target and handle objects was smaller than 10 centimeters. Both modes were controlled by separate buttons. A third button was used for alignment confirmation. After pressing-down this button, the interaction stopped, and the pose of the handle object was fixed. As a mechanism to avoid registering false positives, this button had to be continuously pressed for two seconds.

4.2.2 Tutorial

To train the participants on how to manipulate the handle objects, users were asked to complete a tutorial session in which they interacted with the objects and the virtual helpers. The objects used for the tutorial were not used during the formal study. The three virtual helpers presented in Subsection 3.3, as well as an additional scene without any helper, were introduced in this stage. Participants were able to freely interact with the objects and helpers until they felt comfortable manipulating them.

4.2.3 Alignment Task

For the performance evaluation, users were asked to align as best as possible a handle object with respect to the pose indicated by a target object. Participants were allowed to take as much time as needed to achieve the task. A total of 16 different alignment trials were presented to the users, as a result of the possible combinations of objects (Fig. 2) and virtual helpers (Fig. 1), including one scenario without helpers that served as baseline. One pair of handle and target objects was shown per trial and these were automatically replaced after alignment confirmation from the user.

4.2.4 Eye Tracking Data

To collect information regarding user's focus of attention during alignment, we used the built-in eye tracking system of the HTC Vive. The information corresponding to the three-dimensional points where the users focused while looking at the handle objects and the virtual helpers was computed and recorded by intersecting the scene geometries with the gaze direction.

4.2.5 System Usability Score and Mental Effort

After completing all of the alignment tasks, participants were asked to evaluate the virtual helpers by using the system usability score (SUS) questionnaire introduced by [2], as well as to report the mental effort (ME) perceived based on the 9-point scale proposed by [26].

4.3 Parameter Selection

Through a pilot study, we determined the optimal settings for the experiment. All virtual objects were placed at a height of 1.35 meters. Target objects were placed in the center of the scene with the same orientation, and the handle objects appeared with a pseudo-random orientation and 0.5 meters distance from the target with respect to the World-Right vector. The virtual helpers were placed at 0.5 meters with an angle of 150° between the Handle-to-Target and Target-to-Helper vectors. These parameters are illustrated in Fig. 3. To mitigate possible learning effects, a 16×16 Latin square matrix was used to define the order of the combinations of helper and model presented to participants.

4.4 Experimental Variables

Two independent variables, each one with four different levels, corresponding to the virtual helper and model type were considered in our study. Data regarding the poses of the controller, user's head, and handle and target objects, as well as task duration, user's gaze, and post-tests were used to compute nine dependent variables: a) Position errors computed as the Euclidean distance between the gravity centers of the handle and target objects. b) Orientation errors between the target and handle objects in degrees using an axis-angle representation. c) Time to completion considered as the time elapsed since the first change in the pose of the handle object until user confirmation. d) Users interaction computed as the number of normal and precise manipulations used during trials. e) User's focus of attention determined as the time spent by the user while looking to the helpers and handle objects. f,g) User's distance traveled and average head velocity using the head pose and time to completion. h,i) Usability and mental effort scores.

4.5 Hypotheses

Our general assumption is that the presence of the additional views provided by the helpers will improve the user's performance and experience, especially in scenarios where the user's movements are restricted by the environment (e.g. presence of obstacles such as furniture or equipment). Therefore, we expect the following hypotheses to be true:

H1. Users perform significantly better in terms of position and orientation when virtual helpers are present in the scene.

H2. Users complete the alignment task significantly faster when using virtual helpers.

H3. The presence of virtual helpers significantly reduces the distance traveled and the average head velocity of the users.

H4. The presence of virtual helpers does not significantly affect user's interaction during task performance.

H5. The presence of virtual helpers significantly increases the usability, and reduces the mental effort perceived by users.

5 RESULTS

Due to the non-normal distribution observed in the data collected and the within-subject design of the study, we used Friedman tests with $\alpha = 0.05$ to compare our experimental variables. Posterior Dunn-Sidak tests were used to reveal any statistical significance. Such differences are represented as $\star (p < 0.05)$, $\star\star (p < 0.01)$, and $\star\star\star (p < 0.001)$ in the violin plots [10]⁴ (e.g. Fig. 5).

Since the aim of this study is to demonstrate the feasibility of using the virtual helpers during alignment tasks, rather than to understand how the model selection influences users' performance, the violin plots shown in this Section correspond only to the results obtained grouped by virtual helper. Nevertheless, considering this information can result of special interest for the reader, the statistical results obtained by grouping the data acquired by model are also included in this Section⁵.

5.1 Position, Orientation, and Time to Completion

To evaluate the quantitative results collected for alignment accuracy and time to completion, we grouped the data based on the type of virtual helper used during alignment. The lowest median values for position, rotation and time to completion were reported by the scenarios without the presence of virtual helpers. Friedman tests did not reveal statistical significance between virtual helpers neither for position nor orientation, however statistically significant differences for time to completion ($\chi^2(3) = 16.19$, $p < 0.01$) were found. Participants were able to align the objects without using virtual helpers ($Mdn = 48.66sec$, $IQR = 42.36$) significantly faster when compared to the Top-View Camera ($Mdn = 58.35sec$, $IQR = 44.14$, $p < 0.01$) and the Dynamic Mirror ($Mdn = 60.80sec$, $IQR = 43.30$, $p < 0.01$). These results are summarized and shown in Fig. 5.

When comparing the obtained scores by models, Friedman tests revealed significant interaction for position ($\chi^2(3) = 29.33$, $p < 0.001$), orientation ($\chi^2(3) = 118.04$, $p < 0.001$), and time to completion ($\chi^2(3) = 63.7$, $p < 0.001$). Participants achieved significantly better position scores for the Mug ($Mdn = 2.2mm$, $IQR = 1.6$) than for the Skull ($Mdn = 2.8mm$, $IQR = 2.9$, $p < 0.001$), as well as when aligning the Camera ($Mdn = 1.9mm$, $IQR = 1.8$) compared to the Skull ($p < 0.001$). Dunn-Sidak tests for orientation error showed strongest statistical significance when aligning both, the Balloon ($Mdn = 1.20^\circ$, $IQR = 1.26$) and the Camera ($Mdn = 1.56^\circ$, $IQR = 1.46$), compared to the Skull ($Mdn = 2.75^\circ$, $IQR = 2.67$, $p < 0.001$) and the Mug ($Mdn = 2.66^\circ$, $IQR = 2.10$, $p < 0.001$). Similarly, users were able to align the Mug ($Mdn = 42.99sec$, $IQR = 29.52$) significantly faster than the Skull ($Mdn = 55.26sec$, $IQR = 40.55$, $p < 0.01$) and the Balloon ($Mdn = 74.86sec$, $IQR = 47.83$, $p < 0.001$); the Camera ($Mdn = 53.21sec$, $IQR = 37.22$) when compared to the Balloon ($p < 0.001$); and the Skull than the Balloon ($p < 0.001$).

⁴The dots represent samples and the area surrounding them depicts the data distribution. The white dots in the middle indicate the median while the bottom and top edges of the gray boxes indicate the 25th and 75th percentiles.

⁵A complete report of the results grouped by model, including the violin plots is available as supplementary material.

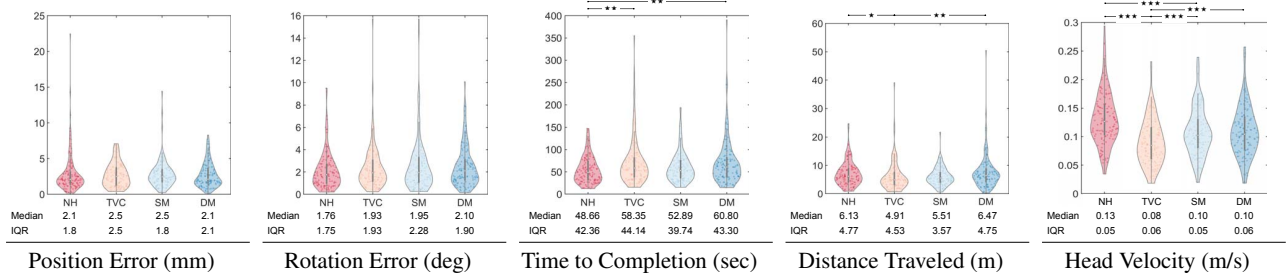


Figure 5: Scores obtained by users under test conditions. Results grouped by virtual helpers: NH-No Helpers, TVC-Top-View Camera, SM-Static Mirror, DM-Dynamic Mirror.

5.2 Normalized Distance Traveled and Head Velocity

To evaluate if the use of these techniques reduces the distance traveled by the users, frequently related to motion parallax, we collected the user's head pose and we used it to compute the total distance traveled for every alignment trial. This information was grouped by model and helper type. Friedman results revealed statistical significance ($\chi^2(3) = 15.27, p < 0.01$) between virtual helpers. Posterior Dunn-Sidak tests shown that users significantly traveled lower distances using the Top-View Camera ($Mdn = 4.91m, IQR = 4.53$) when compared to the scenarios without virtual helpers ($p < 0.05, Mdn = 6.13m, IQR = 4.77$) and with the dynamic mirror ($p < 0.01, Mdn = 6.47m, IQR = 4.75$). As a result of grouping the data by model ($\chi^2(3) = 35.99, p < 0.001$), the distance traveled while aligning the Mug ($Mdn = 5.20m, IQR = 3.92$) and the Camera ($Mdn = 5.01m, IQR = 3.67$) was significantly lower than the one required to align the Balloon ($Mdn = 7.19m, IQR = 5.01, p < 0.001$ for both), as well as when aligning the Camera when compared to the Skull ($Mdn = 6.12m, IQR = 5.12, p < 0.05$) and the Balloon ($p < 0.001$).

Additionally, we computed the average velocity through the alignment task as the ratio of total distance traveled and trial duration. Our statistical analysis revealed significant differences between virtual helpers ($\chi^2(3) = 98.73, p < 0.001$) presenting higher velocity the scenarios without virtual helpers ($Mdn = 0.13m/s, IQR = 0.05$) when compared to all the virtual helpers ($p < 0.001$), while the Top-View Camera ($Mdn = 0.08m/s, IQR = 0.06$) registered the lowest velocity when compared to both of the proposed mirrors ($p < 0.001$). These results can be seen in Fig. 5. On the other hand, statistical significance was found between models ($\chi^2(3) = 37.3, p < 0.001$) presenting the Camera ($Mdn = 0.10m/s, IQR = 0.06$) and Balloon ($Mdn = 0.10m/s, IQR = 0.05$) significantly lower velocities than the Mug ($Mdn = 0.12m/s, IQR = 0.07, p < 0.001$), and the Balloon compared to the Skull ($Mdn = 0.11m/s, IQR = 0.05, p < 0.01$).

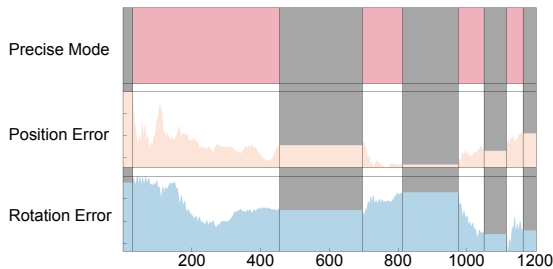


Figure 6: Data sequence corresponding to User2 while aligning the Camera model using the Dynamic Mirror. This sequence illustrates 1200 samples of data collected corresponding to: a) Precise mode manipulation event (top). b) Euclidean distance from target position (middle). c) Angular error from desired orientation (bottom). The gray shaded areas indicate no interaction; thus, the respective errors remain unchanged.

5.3 User Interaction

To investigate if the model or helper involved during the alignment influences users' interaction, we collected the amount of normal and precise manipulation events occurred across every alignment trial. A manipulation event is considered as the press-release action over one of the two interaction buttons of the VR controller used to modify the handle object pose, independently of the duration of the action. An example of how these values were computed can be shown in Fig. 6, where four precise manipulation events can be identified.

After collecting the normal and precise manipulation events, these were grouped by model and virtual helper, similarly to the procedure followed by Subsections 5.1 and 5.5. Friedman tests did not reveal statistical significance when the manipulation events are grouped by virtual helpers (see Fig. 7). However, our tests revealed statistical significance when grouped by model for normal ($\chi^2(3) = 27.71, p < .001$) and precise ($\chi^2(3) = 29.52, p < .001$) modes. Users required a significantly higher amount of normal manipulation events during the alignment of the Balloon ($Mdn = 6.0, IQR = 7.0$) when compared to the Mug ($Mdn = 5.0, IQR = 4.0, p < 0.001$), the Camera ($Mdn = 5.0, IQR = 5.0, p < 0.001$), and the Skull ($Mdn = 5.0, IQR = 5.0, p < 0.001$), and a significantly higher amount of precise manipulations to align the Balloon ($Mdn = 15.0, IQR = 20.0$) when compared to the Mug ($Mdn = 8.0, IQR = 11.5, p < 0.001$), the Camera ($Mdn = 10.5, IQR = 16.0, p < 0.05$), and the Skull ($Mdn = 11.0, IQR = 18.5, p < 0.05$).

To evaluate if the use of the interaction modes improves the overall alignment, we detect the manipulation events for every trial, and we classify them based on three categories. We ranked any normal manipulation event $-NM-$, as $NM++$ if the manipulation event lead to an improvement in position and orientation, as $NM+$ if the manipulation event resulted in an improvement in position but negatively affected the orientation, or vice versa, and as $NM-$ if the interaction event negatively affected both, position and orientation. The individual scores for these subsets were divided by the total amount of normal manipulation events, helping to normalize the data sample based on the specific alignment trial.

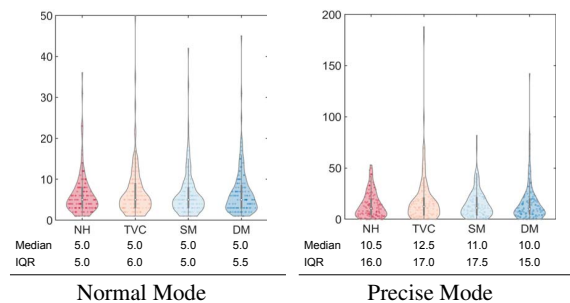


Figure 7: Normal and precise manipulation events grouped by virtual helper. NH-No Helpers, TVC-Top-View Camera, SM-Static Mirror, DM-Dynamic Mirror.

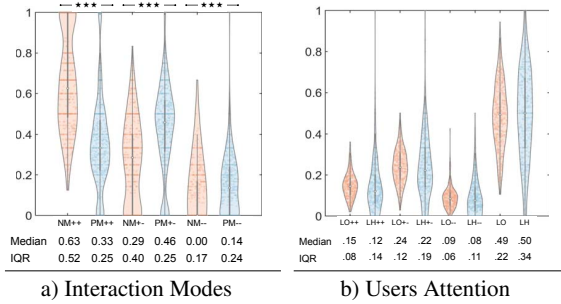


Figure 8: Overall improvement achieved by users during alignment. a) NM-Normal Mode Interaction, PM-Precise Mode Interaction, '++'-Interaction improving both, translation and rotation, '+-'-Interaction improving either, translation or rotation, '-'-Interaction deterioration for both, translation and rotation. b) LO-User looking at handle object, LH-User looking at virtual helper.

The same classification procedure was used to compute the scores for the precise mode. An example of this classification can be seen in Fig. 6 where, the first manipulation event classifies as a PM++ event, the second and third as PM+- while the fourth one as PM-. Friedman tests revealed significant differences between the improvement observed based on the interaction mode used ($\chi^2(5) = 1196.97$, $p < .001$). Users were able to significantly improve the position and orientation of the held objects more frequently while using the normal mode ($Mdn = 0.63$, $IQR = 0.52$) when compared with the precise mode ($Mdn = 0.33$, $IQR = 0.25$, $p < 0.001$) and negatively affected the position and orientation of the objects using the normal mode ($Mdn = 0.00$, $IQR = 0.17$) with significant less frequency than the precise mode ($Mdn = 0.14$, $IQR = 0.24$, $p < 0.001$). Despite statistical significance between all six groups depicted in Fig. 8a was found, only those corresponding to the same type of improvement classification are shown.

5.4 Users Attention

To understand how the presence of virtual helpers in the scene can influence user's attention, and similarly to the approach used in Subsection 5.3, we used the eye tracking data to identify the time frame on which the users were looking at the objects and virtual helpers. We compared if the alignment scores of the handle object improved during this action and classified them based on the three categories used for interactions (i.e. "++", "+-", and "-"). In addition to these classes, we included a fourth class to indicate users looking at the object (LO) or at the helper (LH) without manipulation. Friedman test did not reveal statistical significance between any of these classes when compared against its simile. Nevertheless, it is visible from Fig. 8 that the distribution observed when the helpers are used during alignment is more spread. This suggests that overall, the users spent a similar amount of time solely observing the objects than the time spent observing the virtual helpers. However, when observing this behavior individually, some users spent more time visualizing the virtual helpers than the actual objects and vice versa.

Additionally, the data collected from the eye tracking was used to identify the structures of the handle objects where the users focus their attention during the alignment. To visualize this data, we gathered the 3D points of the object's surface where the users focused their attention during alignment. These points were mapped to their respective UV Map and used to generate a heat map. To increase visibility and to mitigate possible inaccuracies in the values acquired from the tracking system, we used a bi-dimensional normal distribution centered in the position of the points collected. The cumulative UV Maps generated are depicted in Fig. 9.

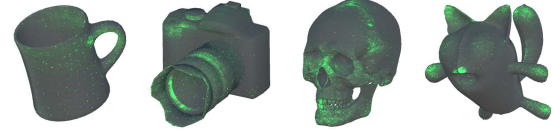


Figure 9: Model-based UV maps generated from users' attention during alignment tasks. This data is retrieved from eye tracking values collected during task performance.

5.5 Usability and Mental Effort

Results collected for usability (SUS) and mental effort (ME) perceived by users indicated that users prefer scenarios with the presence of virtual helpers. As can be seen in Fig. 10, the Dynamic Mirror obtained the best results for both of these metrics, followed by the Static Mirror and the Top-View Camera. The three virtual helpers obtained SUS scores considered above the average, while the scenario without helpers scored below the average⁶. Friedman results revealed significant differences between virtual helpers for usability ($\chi^2(3) = 35.16$, $p < .001$). A Dunn-Sidak post-hoc test revealed that SUS scores were significantly better for the Top-Camera View ($Mdn = 72.50$, $IQR = 28.75$), Static Mirror ($Mdn = 75.00$, $IQR = 21.25$), and Dynamic Mirror ($Mdn = 76.25$, $IQR = 26.25$) compared to the scenes without virtual helpers ($Mdn = 50.00$, $IQR = 25.00$, $p < 0.001$). Similar to the usability scores, Dunn-Sidak results for ME revealed statistical significance between virtual helpers ($\chi^2(3) = 16.97$, $p < .001$). A post-hoc test revealed that the mental effort using the Static ($Mdn = 4.5$, $IQR = 2.5$, $p < 0.01$), and Dynamic ($Mdn = 4.0$, $IQR = 2.5$, $p < 0.001$) mirrors was significantly lower when compared to the scenario without helpers ($Mdn = 7.0$, $IQR = 3.0$).

6 DISCUSSION

This section presents a detailed discussion of the results obtained from the statistical analysis performed to the data collected during our user study. Results from position showed overall median errors no larger than 2.5 mm for the scenarios with and without virtual helpers. Nevertheless, the Top-View Camera and Dynamic Mirror presented the most consistent values. Users were able to achieve alignment under 10 mm for all the alignment trials when using these helpers. In terms of orientation, all scenarios registered overall scores with median error values under 2.5° , and under 10° for most of the individual trials. Results from our study revealed statistical significance for time to completion scores. The scenarios without virtual helpers registered lower values than the Top-View Camera and Dynamic Mirror. When comparing the results obtained against those presented in [23], our study revealed a similar strong

⁶Scores between 68 and 80.3 are considered above average, while scores between 51 and 68 are considered average.

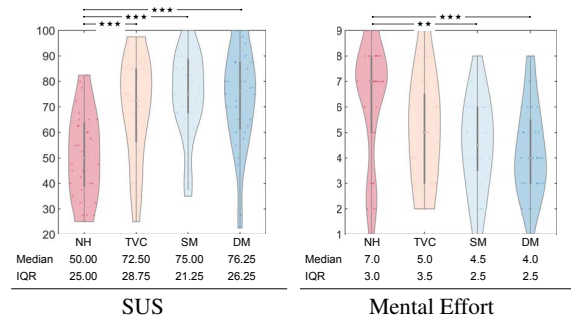


Figure 10: Scores for SUS and ME reported by users. NH-No Helpers, TVC-Top-View Camera, SM-Static Mirror, DM-Dynamic Mirror.

influence of the model type over the position, orientation, and time to completion scores. Nevertheless, it is worth to mention that the scores for time to completion in our study are lower.

Even though we expected users would achieve better alignment –H1– in less time when using the virtual helpers –H2–, the results did not exhibit such a trend. These results are attributed to several factors. According to Buhrmann [3], performing complex tasks requires the coordination of multiple sensorimotor patterns involving actions and perceptual systems that are improved based on the user's experience. Therefore, the lack of familiarity using the additional views for alignment tasks other than for visualization purposes can affect task performance when the helpers are used. Another factor can be associated to the divided attention nature of the task as described by Wickens et al. [36]. This concept describes our limited ability to simultaneously perform multiple concurrent tasks. While an optimal scenario will require users to parallel process the visual input from the direct view and the one provided by the helper, physiological aspects such as the amount of visual area that can be covered by the human eye limits our ability to achieve this. Furthermore, the increasing time observed in the scenarios with virtual helpers can be associated with the lack of familiarity from the users. We expect the users to accept the task of alignment more naturally with no additional support; however, using the virtual helpers induces an extra learning curve that the users do not get the opportunity to adjust in a short study. This can also explain why the time to completion scores for the scenario without helpers and the static mirror are comparable, as humans are accustomed to the use of mirrors.

When comparing the distance traveled by users, the Top-View Camera obtained comparable scores to the Static Mirror but significantly smaller when compared to the Dynamic Mirror and the scenes without virtual helpers. This suggests that using camera views or static mirrors could benefit the alignment tasks in environments where physical obstacles exist. The increase in the distance traveled under scenarios without helpers can be result of motion parallax to generate significant information for alignment, while the increase in the Dynamic Mirror scores are result of users moving around the object to generate reflections from multiple viewpoints. Results obtained from the average head velocity highlight the effects described by the distance traveled, presenting the Top-View Camera the lower velocity values when compared to all the other scenarios, while the scenario without helpers presented significantly higher values than the mirrors. These results support –H3–, and therefore our recommendation to use virtual cameras and static mirrors for scenarios with space constraints.

In terms of interaction, results suggest that the model selection plays a stronger role on the amount of normal or precise manipulation events during alignment, while the virtual helpers do not significantly influence this metric, supporting –H4–. Our study also suggests that users can improve simultaneously the position and orientation of the handle object more frequently using the normal mode. However, this can be influenced by the fact that the precise mode is enabled only when the virtual objects are closer than 10 centimeters. Differently, statistical results derived from user's attention while aligning the objects showed comparable performance improvement when looking at the handle object or when looking at the virtual helpers. Furthermore, based on the observation of the heat maps generated by using the gaze data collected, users seem to focus their attention in surface areas with high curvature such as the camera's lens and shooter button, the bones close to the eyes, nose and jaw of the skull, or the balloon's mouth. Similarly, this data suggest that the salient geometrical structures of the objects are also used for alignment. Example of this are the mug's handle, the camera's lens, and the balloon's legs, hears and tail.

In terms of usability and mental effort, our study revealed significantly higher SUS scores, as well as significantly lower mental effort when using virtual helpers, supporting –H5–. Interestingly, despite

that no statistical significance was found for alignment scores, and even though users were able to align significantly faster the objects without helpers; the SUS and ME scores reflect a clear preference for the presence of the virtual helpers. Additionally, an interesting phenomenon is observed for the ME scores without helpers, where a small subset of users reported very low scores, while the rest of the sample reported high scores, showing a clear divided opinion regarding the mental effort perceived.

7 CONCLUSIONS & FUTURE WORK

This paper presents the results of a user study as the first step towards exploring the benefits of using additional views generated from virtual cameras and mirrors to assist users during alignment tasks. Furthermore, to our knowledge, this is the first work that studies the potential use of mirrors with true parallax properties for alignment tasks in Mixed Reality environments.

Overall results show that users were able achieve overall errors under 2.5 mm for position and 2° for orientation when using virtual helpers. Time to completion scores reflected comparable scores between the scenarios with static mirrors and the scenarios with not helper, being the last ones, significantly faster than the scenarios with top-view cameras and dynamic mirrors. On the other hand, analysis in terms of model behavior revealed the highest scores for position when the Camera and the Mug models were used, while the Balloon and Camera models present the best scores for orientation. In terms of time to completion, the Mug and Camera models were the fastest model to align, while the Balloon was the slowest.

Based on the scores obtained for user's traveled distance and average head velocity, our results showed that using top-down camera views and static mirrors benefit alignment for scenarios where users displacement is limited, such as those scenarios observed in medical and industrial environments. On the other hand, dynamic mirrors appear to benefit scenarios that require users to visualize the objects from multiple perspectives such as visualization and exploration tasks. Therefore, this type of helper is recommended for scenarios where users can freely move around the scene.

Users' preference significantly favors the use of virtual helpers as reflected by the usability and mental effort scores collected. These results show significantly higher usability scores and lower mental effort when the virtual helpers are used to assist the user during task performance. This indicates that the virtual helpers are suitable to assist users during alignment tasks.

7.1 On Transferring Virtual Helpers to AR Environments

Based on the performance scores achieved by using the virtual helpers proposed in this work, and as a part of our future work, we expect to extend our user study and analyze their performance when virtual and real objects interact in AR environments.

The Top-View Camera. We expect that the top-view can be successfully transferred from virtual to augmented reality environments by integrating external cameras placed on top of the scene. In this scenario, the video feed from an external camera used to provide the top-down view can be streamed to an AR headset and visualized by the user while performing the alignment task. Existing work presented by [28] has used a similar approach to visualize a dense point cloud acquired from a binocular endoscope using an AR optical see-through head-mounted display.

The Static Mirror. Similarly to the top-view camera, transferring the static mirror into AR environments is possible. In this case, generating the reflections of the virtual and real content of an AR scene can be done by the integration of real mirrors and AR headsets. To do so, it is necessary to track the pose of the mirror from the user's viewpoint. This pose can later be used to define the mirror's surface at which the reflection of the virtual content can be generated. Hence, the physical properties of the mirror would allow

to observe the reflection of the real content, while the poses of the mirror and the user's viewpoint can be used to generate the reflection of the virtual content. A similar idea, has been suggested by [38] in which, the position and orientation of a non-movable mirror is used to define the pose of a virtual plane able to generate the reflections of the virtual content observed in an AR environment.

The Dynamic Mirror. Moreover, and even though the integration of a dynamic mirror can be more complex than the integration of the other virtual helpers, this can be achieved by attaching the mirror to an adjustable holder. In this regard, and following the working principle described for the static mirror, the adjustable holder could be able to benefit from the user's viewpoint to automatically adjust the position of a physical mirror.

7.2 On Exploring Additional Features

In addition to the integration of the virtual helpers into AR environments, we hope that the eye-tracking data collected from this study can be used to improve user's comfort and ergonomics when using our techniques, as well as to explore the feasibility of automatically determining the optimal position of the virtual helpers based on the geometry of the objects to align.

Even though in this work we focused on investigating how the virtual helpers influence the user's while aligning same sized objects, which generally is the case of real-to-virtual alignment, we believe it would be particularly interesting to explore additional object's properties such as changes in scale, that covers the case of virtual-to-real alignment.

Moreover, whereas our study considered a single helper size and type as a means to generalize our method and avoid the addition of independent variables that are not directly related to the helper type, we are also interested on investigating if using different sizes or instances of the virtual helpers, either of the same or different types, may provide additional benefits to the user's during the alignment task. Lastly, if tracking of the object of interest is available, we would like to investigate how the use of the helpers can be compared against traditional guidance techniques; and eventually, even adapt the position and orientation of the virtual helpers based on the alignment error observed between the handle and target objects.

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